```
                    SUPERVISED LEARNING MODELS


import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt
iris = load_iris()
X = iris.data
y = iris.target
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.3,
random_state=42)
svm_model = SVC(kernel="linear")
svm_model.fit(X_train, y_train)
y_pred_svm = svm_model.predict(X_test)
print("SVM Accuracy:", accuracy_score(y_test, y_pred_svm))
print("SVM Confusion Matrix:\n", confusion_matrix(y_test, y_pred_svm))
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)
y_pred_rf = rf_model.predict(X_test)
print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
print("Random Forest Confusion Matrix:\n", confusion_matrix(y_test, y_pred_rf))
plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
sns.heatmap(confusion_matrix(y_test, y_pred_svm), annot=True, cmap="Blues",
fmt="d")
plt.title("SVM Confusion Matrix")
plt.subplot(1, 2, 2)
sns.heatmap(confusion_matrix(y_test, y_pred_rf), annot=True, cmap="Greens",
fmt="d")
plt.title("Random Forest Confusion Matrix")
plt.tight_layout()
plt.show()
```

Output:

```
SVM Accuracy: 0.9777777777777777
SVM Confusion Matrix:
 [[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]
Random Forest Accuracy: 1.0
Random Forest Confusion Matrix:
 [[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

